Homework 4: Test for Stutter

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Course: Secure Software Testing

Program File	Stutter.js
GitHub Repository	https://github.com/tainyu/secure-software-testing-2025-spring-classroom-lab-2-114-spring-secure-software-testing-1/tree/main/HW4

This report answers parts (a)-(d) of Homework 4 based on the supplied Stutter.js program and follows the submission guideline requiring a GitHub link and named report file.

(a) Control Flow Graphs

The following CFGs are drawn for all functions defined in Stutter.js: stut(inFile), checkDupes(line), isDelimit(C), and main().

圖 1- CFG_stut

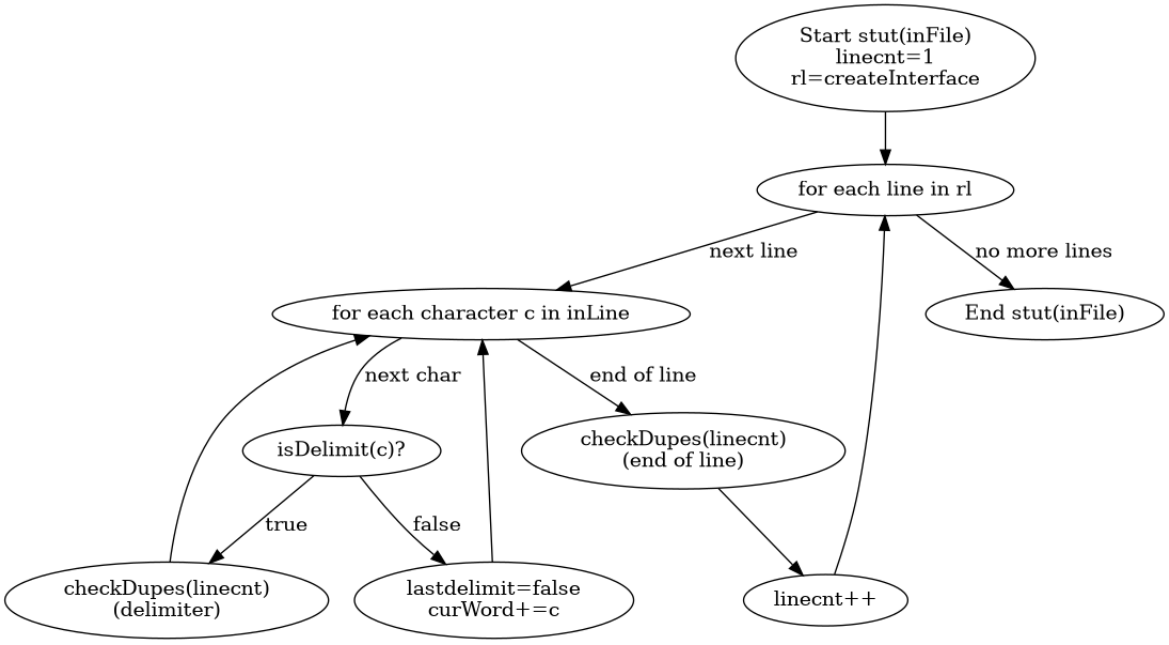


圖 2- CFG_ checkDupes

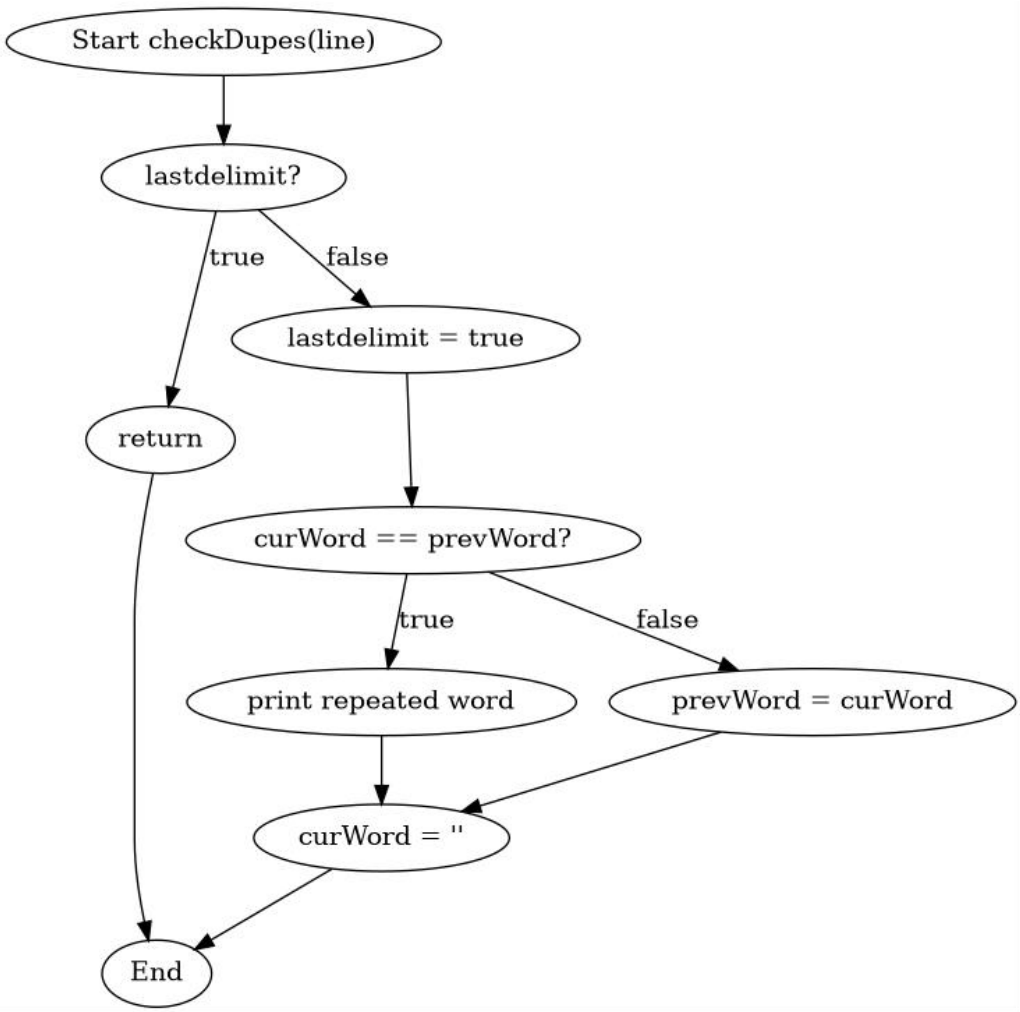


圖 3-CFG_isDelimit

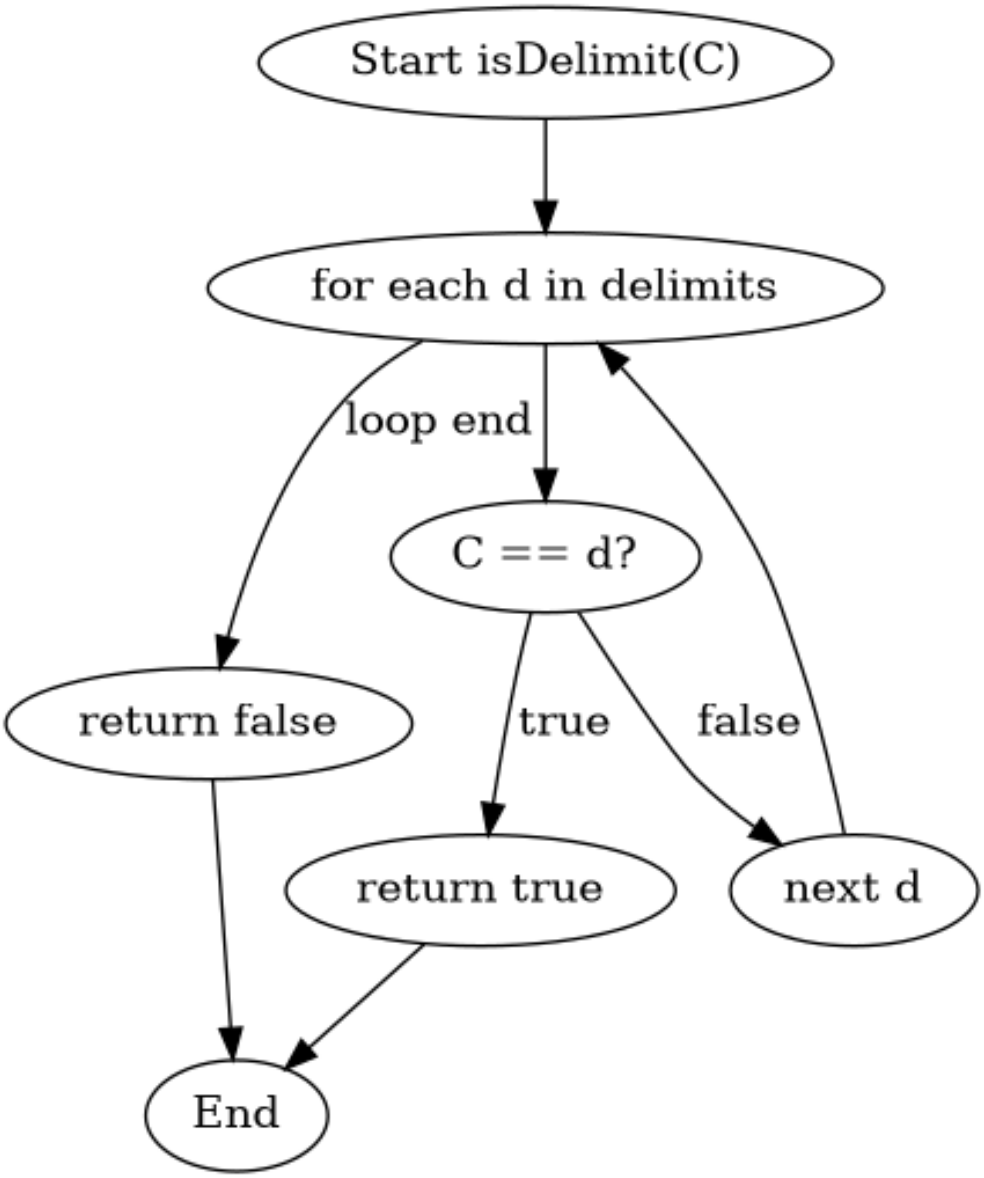
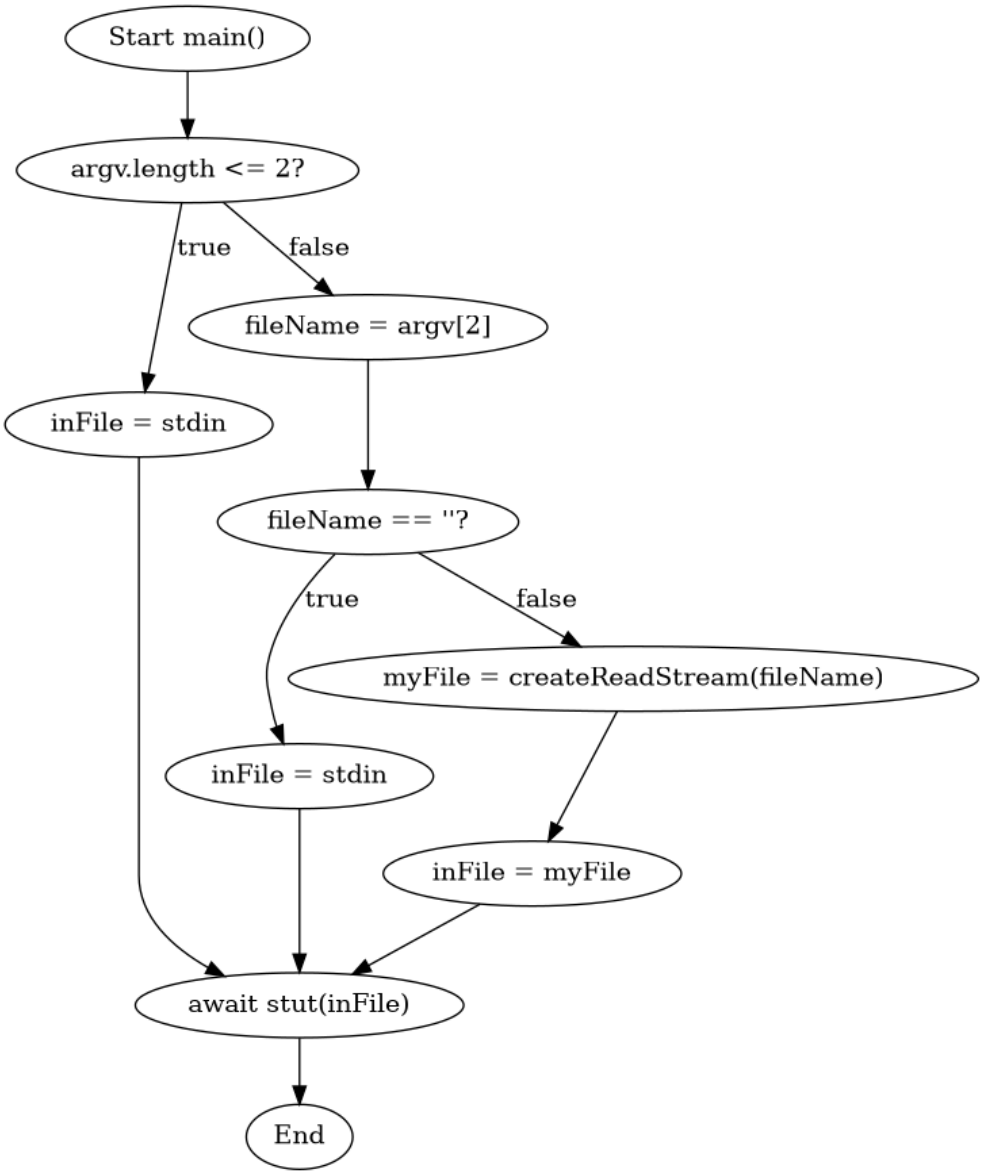


圖 3- CFG_main


Start main()
argv.length <= 2?
true
false
fileName = argv[2]
inFile = stdin
fileName == ''?
true
false
myFile = createReadStream(fileName)
inFile = stdin
inFile = myFile
await stut(inFile)
End

(b) Call Sites of Functions Defined in Stutter.js

The table below lists every call site whose callee is a function defined inside Stutter.js. Calls to library functions such as fs.createReadStream() are excluded because they are not functions defined in Stutter.js.

No.	Caller	Call Site	Callee	Notes
1	stut(inFile)	if (this.#isDelimit(c))	isDelimit(C)	Called once for each character in a line.
2	stut(inFile)	this.#checkDupes(linecnt) inside delimiter branch	checkDupes(line)	Triggered when a delimiter ends a word before end-of-line.
3	stut(inFile)	this.#checkDupes(linecnt) after inner loop	checkDupes(line)	Triggered at end of each input line.
4	main()	await Stutter.stut(inFile)	stut(inFile)	The selected input stream is passed to stut().
5	module entry	if (require.main === module) { main(); }	main()	Program-start call site; no actual parameters are passed.

(c) Coupling DU-Pairs for Each Call Site in (b)

A coupling DU-pair connects an actual parameter definition in the caller to a reachable use of the corresponding formal parameter in the callee. For the module-entry call to main(), there is no parameter passing, so no coupling DU-pair exists.

No.	Call Site	Actual-In Def	Formal-In	Reachable Use in Callee	Result
1	stut() -> isDelimit(c)	c defined by: for (const c of inLine)	C	Used in decision: if (C === d)	DU-pair exists
2	stut() -> checkDupes(linecnt) [delimiter call site]	linecnt defined by initialization linecnt = 1 or by linecnt++ from earlier iterations	line	Used in output: Repeated word on line ${line} ...	DU-pair exists
3	stut() -> checkDupes(linecnt) [end-of-line call site]	linecnt defined by initialization linecnt = 1 or by linecnt++ from earlier iterations	line	Used in output: Repeated word on line ${line} ...	DU-pair exists
4	main() -> stut(inFile) [argv.length <= 2 path]	inFile = stdin	inFile	Used in readline.createInterface({ input: inFile, ... })	DU-pair exists
5	main() -> stut(inFile) [fileName === "" path]	inFile = stdin	inFile	Used in readline.createInterface({ input: inFile, ... })	DU-pair exists

No.	Call Site	Actual-In Def	Formal-In	Reachable Use in Callee	Result
6	main() -> stut(inFile) [valid file path]	myFile = fs.createReadStream(fileName); inFile = myFile	inFile	Used in readline.createInterface({ input: inFile, ... })	DU-pair exists
7	module entry -> main()	No actual parameter	None	None	No coupling DU-pair

(d) Test Data for All-Coupling-Uses Coverage

The following test set covers all coupling uses identified above. The key idea is to force each actual-parameter definition to reach at least one real use in the callee. For checkDupes(line), the formal parameter line is only used when a repeated word is reported, so the test data must actually trigger the duplicate-word message.

Test	Execution Mode	Input / File Content	Coverage Purpose	Expected Result	AUC Covered
T1	node Stutter.js (stdin path)	STDIN: "go go " (with trailing space)	Covers stut()->isDelimit(c) with both non-delimiter and delimiter characters; covers delimiter call site stut()->checkDupes(linecnt); covers main()->stut(inFile) where inFile = stdin via argv.length <= 2.	Program prints: Repeated word on line 1: go go	1, 2, 4
T2	node Stutter.js (stdin path)	STDIN: "abc\nabc"	Covers end-of-line call site stut()->checkDupes(linecnt); linecnt reaches line use when duplicate is detected on line 2.	Program prints: Repeated word on line 2: abc abc	3
T3	node Stutter.js "" (empty filename path)	STDIN: "x x " (with trailing space)	Covers main()->stut(inFile) where argv[2] is the empty string and therefore inFile = stdin through the second definition site in main().	Program prints: Repeated word on line 1: x x	5
T4	node Stutter.js sample.txt (valid file path)	sample.txt content: "red red " (with trailing space)	Covers main()->stut(inFile) where inFile is defined from a file stream created by createReadStream(fileName).	Program prints: Repeated word on line 1: red red	6

Conclusion: T1-T4 collectively satisfy All-Coupling-Uses Coverage for the coupling DU-pairs in part (c). The module-entry call to main() does not contribute a coupling DU-pair because no parameter is passed.